

TECH IS THEIR TRADE
The scene at a research lab in IIT Delhi



"Institutions must see a shift in pedagogy so as to foster an interest in STEM subjects. Learning at Deeksha STEM revolves around problem-solving, self-research, decision-making, innovation and communication"

– DR SRIDHAR G.
Founder, Deeksha STEM School

the global leader in terms of absolute numbers. However, there is an urgent need to expand the STEM universe. The Union government has encouraged STEM by introducing initiatives such as the Atal Tinkering Labs, Dhruv Scheme and Rashtriya Avishkar Abhiyan at the school level. Programmes like Make in India, Digital India, Start-up India and Skill India are expected to deepen the pool. However, India needs to put in more money in STEM education. The Science and Technology Policy in 2003 mandated that 2 per cent of India's GDP would be spent on research and development by 2007. This goal has yet to be achieved.

The UNESCO Science Report of 2021 says India's average GERD (gross domestic expenditure on R&D) has been just 0.75 per cent of its GDP in the past two decades. It's the lowest GERD-GDP ratio among the BRICS nations: Brazil, Russia, India, China and South Africa. Though 43 per cent of all Indian STEM graduates are women, their share in STEM jobs is a minuscule 14 per cent—this reflects a global asymmetry that India must break. The National Education Policy 2023 has placed emphasis on STEM, but not only does it require speedy implementation, it must also be backed by adequate funding. Only then will India own the future. ■

THE ATAL TINKERING LABS (ATLS)

They are aimed at becoming hubs of innovation for school students. Over 10,000 ATLs have been established. ATLs are being nurtured by top organisations. Indian Space Research Organisation has adopted 100 ATLs. OPPO India will set up an ATL in Kerala—the first public-private collaboration for ATLs

DEEKSHA STEM SCHOOLS

Uniting conventional education with STEM and child well-being, Deeksha schools encourage self-learning and innovation